Band-1 Hadamard Order 9104 – Classical Paley-I (q=9103); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9104 Paley-I PASS; Paley-II SKIP (q=4551=3*37*41 NOT prime power); Kronecker SKIP (prefer Paley-I); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-21

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-21_Band1_Hadamard_9124_PaleyII`; `2026-09-21_Band1_Hadamard_9092_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_9104_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9104 only (verified max|HHˆT-nI|=0), via classical Paley-I q=9103 (prime ===3 mod 4). 9104 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9104: Paley-I **PASS**s ($q = 9103$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL** ($q = 4551 = 3 \cdot 37 \cdot 41$ NOT prime power); Kronecker construction **SKIP**ped this wave (prefer Paley-I; note $9104 = 4 \cdot 2276$ arithmetically admissible). **9104** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9124+9092+2220+8884+2236+2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **9104**.

- H_{9104} = Paley-I ($q = 9103$) – classical SPK
- Paley-II for 9104: **SKIP** ($q = 4551 = 3 \cdot 37 \cdot 41$ NOT prime power)
- Kronecker for 9104: **SKIP** construction this wave (prefer Paley-I)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_9104.py`

Results. overall_pass=True; units 97/97; wall 614.464s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9104 = q + 1$ with $q = 9103$ prime $\equiv 3 \pmod 4$ -> Paley-I eligible.
- Paley-II: $q = 4551 = 3 \cdot 37 \cdot 41$ NOT prime power -> SKIP.
- Kronecker: $9104 = 4 \cdot 2276$; both Hadamard-admissible; construction SKIPPED this wave (prefer Paley-I).
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9104 Paley-I	Paley-I q=9103	0	606.8747
H_9104 Paley-II	SKIP q=4551=3*37*41 NOT prime power	–	–
H_9104 Kronecker	SKIP prefer Paley-I (4*2276 not built)	–	–

SPK closure note

Seeds include prior SPK through **9124** lineage (**9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; **1064+2128**; ...; **NOT** 8948/8644/8516/...) + verified **9104**. Next constructive gap **2052** (OPEN).

Quantity	Value
9104 in SPK	**True** (Paley-I q=9103)
Gap 2052	**OPEN**
clay_closed	false

Quantity	Value
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=2282->n=9128 Paley-I q=9127 prime ===3 mod4 – alternate SPK OR q=2293->n=9172 triage – Paley-I q=9171 composite; Paley-II q=4585=5 7131 NOT prime power; Kronecker 4*2293 OR n=2052 standard SPK constructive gap – method-attack / lit OR q=2288->n=9152 Paley-I DIRECT q=9151 OR method-attack SPK gap 2052 OR Track B / Object 1 meter await